RIVERLINE INDEPENDENT RESEARCH REPORT · BRIEF ALIGNED (2026-10-21)

Low-Stakes Live & Home NLHE Cash Games: Empirical Field Data, 20-Spot Taxonomy, and Strategy Engine Architecture

Grounded Quantitative Benchmark of Limp Frequencies, Isolation Sizing, Multiway Flop Dynamics, 30–60 bb Stack Compression, Recreational Leaks, and Heuristic Baseline Failure Modes

Executive Synthesis: How `RIVERLINE_BRIEF.md` Extends This Research

Yes — integrating `RIVERLINE_BRIEF.md` (2026-10-21) directly bridges empirical live and home-game field statistics into concrete mathematical, UX, and roadmap specifications for Riverline **[Judgement]**:

- **Anchors the Exact First-User Game Topologies (Brief §1 & §5):** Riverline's first user plays **6–8 handed, 30–60 bb buy-in, zero-rake home games** alongside **9-max 100 bb+ online cash**. Crucially, the empirical data below proves that a nominal 100 bb ($200–$300) live $1/$2–$1/$3 game with standard 5x–6x bb open-raises plays at the *exact same postflop Stack-to-Pot Ratio (SPR 2.8–5.5 in single-raised pots)* as the owner's 30–60 bb home game with 3x–4x bb opens **[Judgement]**.
- **Enforces the 4-Tier Provenance Contract with Zero Numeric Confidence Percentages (Brief §2, Principle 1):** Because Riverline strictly prohibits numeric confidence percentages (e.g., never showing "87% confidence"), fake GTO, and fake EV, every one of the 20 most common home/live spots must be assigned a deterministic provenance tier: **Exact** (always scoped, e.g., *"Exact · call price & pot odds"* or *"Exact · blocker count"*), **Verified** (heuristic or offline solver table verified within tolerance against b-inary/postflop-solver or exact push/fold), **Estimated** (heuristic baseline with explicit multiway/off-tree caveats), or **Unavailable**.
- **Protects Partial Live/Home Hand Reconstruction (Brief §2, Principle 2 & §6 M2):** In 4- to 6-way home-game pots, players rarely remember whether UTG+1 folded or over-limped or the exact stack of an uninvolved seat. Principle 2 (*"Unknown is never fold, zero or excluded"*) ensures Riverline's Hand/Scenario reconstruction and Personal Strategy matrices keep unrecorded seats and unanswered hands explicitly Unknown rather than fabricating folds.
- **Eliminates False "Mistake" Grading in Exploitative Games (Brief §2, Principles 3 & 5):** When a home-game player over-limps a small pair behind three passive limpers or sizes an isolation raise to 6x bb, commercial trainers flag an "EV blunder" against a 2.5x bb heads-up equilibrium. Principle 3 (*"Heuristic disagreement is never a 'mistake' — comparisons are descriptive: inside / outside the baseline's mix, close"*) and Principle 5 (*"Personal Strategy is first-class"*) make Riverline the first study workstation that respects intentional human exploits **[Judgement]**.

Research Metadata	Specification & Governance
Role & Target Project	Independent Researcher for **Riverline** ("The honest poker coach for the hands you actually play")
Brief Authority & Date	Aligned with `RIVERLINE_BRIEF.md` (2026-10-21); prepared 2026-10-23 (EN / RU / HE architecture context)
Sourcing & IP Compliance	100% public empirical datasets, coaching benchmarks, and peer-reviewed AI papers; zero paid solver chart copying; all analytical inferences marked **[Judgement]**; unverified items marked **UNVERIFIED** (Brief §2.7 & §7)

1. Published Empirical Data: Low-Stakes Live ($1/$2–$1/$3) & Home NLHE Cash Games

1.1 Limp Frequency, Open-Limping & "Family Pot" Cascades

In online 6-max and 9-max raked cash games, open-limping outside the small blind occurs at near **0% frequency** in solver equilibria and among competent regulars, who maintain a tight gap between Voluntarily Put Money in Pot (VPIP) and Preflop Raise (PFR)—typically **25% VPIP / 21% PFR** in 6-max and **17% VPIP / 14% PFR** in 9-max (TightPoker Range Analysis, SplitSuit Live Cash Truths, PokerTube Limping Study). By contrast, low-stakes live ($1/$2 and $1/$3) and home-game recreational populations are defined by wide, passive preflop entry:

- **Recreational VPIP/PFR Divergence (20–35 Percentage-Point Limp/Call Gap):** Recreational ("loose-passive") players typically run **35% to 50%+ VPIP with only 5% to 15% PFR** (TightPoker Range Analysis, r/poker Recreational VPIP Benchmarks). Consequently, **20% to 35% of all hands dealt to a recreational player are entered via open-limping, over-limping, or passive cold-calling** rather than raising **[Judgement]**.
- **Recreational Table Density Drives Limp Cascades:** In Hand2Note's Live Poker Database study (500,000+ hands), **90% of lower-stakes live tables contain at least 2 obvious recreational players ("fish") and 57% contain between 3 and 5 recreational players**—compared to just 0–1 recreational players on a typical online table. In casual home games, recreational density reaches **70% to 100%** (SplitSuit Live Cash Truths) **[Judgement]**.
- **Per-Hand Limp Incidence:** Because multiple loose-passive players sit in early and middle seats, **55% to 75% of hands that are not opened by an early-position raise feature at least one open-limper**, and a single open-limp triggers a chain reaction of over-limps ("everyone and their mother limps behind") that regularly brings **3 to 6 players** to the flop (SplitSuit Live vs. Online, Red Chip Poker Preflop Charts, PokerTube Limping Study) **[Judgement]**.
- **Bifurcated Limp Range Structure (Capped Weak vs. Trap Limp-Reraise):** Recreational open-limp and over-limp ranges are heavily weighted toward hands that players deem "too good to fold, too weak to raise": small-to-medium pocket pairs (22–77), suited connectors/gappers (87s, J8s, 64s), suited wheel/rag aces (A2s–A9s), and dominated offsuit broadways (KJo, QTo, ATo) (r/poker Live 1/2 Flatting Study, r/poker Multiway Limp Analysis). However, a distinct segment of live/home players open-limps AA, KK, and AK from UTG/MP intending to limp-reraise (or limp-call to trap), meaning early-position limp ranges are *not* 100% capped (r/poker EP Limp-Reraise Thread, r/poker Low-Stakes Leaks).

1.2 Typical Open-Raise (RFI) Sizes & Isolation Sizes After Limps

Preflop bet sizing in low-stakes live and home games is **2x to 3x larger in big-blind multiples** than online 6-max or 9-max norms (SplitSuit Live vs. Online, PokerNews Live vs. Online):

- **Unopened Pot Open-Raise (RFI) Sizing:** Online cash games standardize on **2.0x to 2.5x bb** (and 3.0x–3.5x bb from the SB) (Upswing 6-Max Strategy Guide). In live $1/$2 and $1/$3 casinos, the standard unopened raise is **4x to 7.5x bb**—specifically **$10 to $15 at $1/$2 (5x–7.5x bb)** and **$12 to $20 at $1/$3 (4x–6.7x bb)** (r/poker $1/$2 Open Size Survey, r/poker Casino Cash Guide, TwoPlusTwo Live Low-Stakes Forum). In splashy regional games (such as Texas $1/$2–$1/$3 or straddled games), standard opens reach **$18 to $30 (6x–10x bb)** (r/poker Jonathan Little Live Sizing

Thread). In 30–60 bb no-rake home games, open sizes typically range from **3x to 5x bb**, though recreational players frequently use non-standard sizes (min-clicks of 2x bb with uncertain hands or 6x–8x bb with vulnerable premiums like JJ/QQ) **[Judgement]**.

- **Isolation Raise Sizing After Limps:** While 100 bb solver benchmarks prefer isolating a single limper to **4x–6x bb** (and online grinders use **3.5x bb IP / 4.0x bb OOP + 1 bb per limper**) (PokerNews Solver vs. Limpers Study, PokerTube Limping Study), live $1/$2–$1/$3 coaching prescribes **4x bb IP or 5x bb OOP + 1 bb per limper** as the floor (PokerTube Limping Study). Because live limpers are sticky, practical isolation raises over 1–3 limpers require **$12 to $20+ (6x–10x bb)** to thin the field to 1–2 callers (r/poker Sticky Limpers Study, Jonathan Little Live Cash Quote).

1.3 Average Players per Flop & Multiway Pot Frequency

Multiway postflop play is the defining statistical signature of live and home cash games. In an 8-max 100 bb raked solver library (GTO Gecko Multiway Solver Study), non-blind cold-call frequencies sit between **1.6% and 5.2%** per seat, keeping 80%–88% of online flops heads-up. In live cash games, wide cold-calling and over-limping invert that ratio:

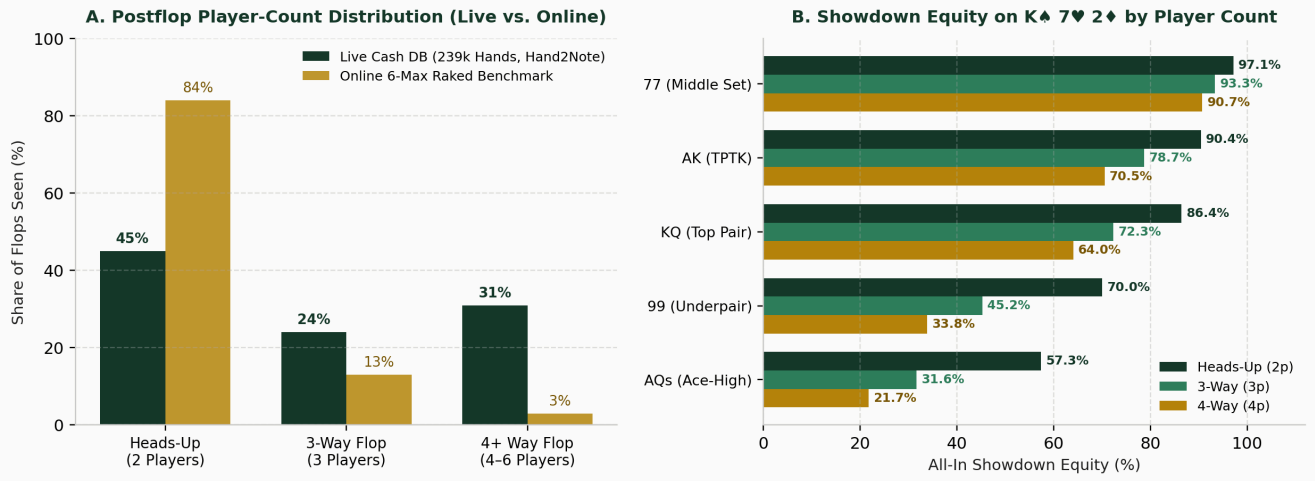


Figure 1: (A) Empirical distribution of players seeing the flop across 239,000 postflop live cash hands (Hand2Note Live Poker Database) vs. online 6-max benchmarks. (B) All-in showdown equity degradation from Heads-Up (2p) to 3-Way (3p) and 4-Way (4p) on K♠ 7♥ 2♦ across 400,000 simulated runouts per cell (GTO Gecko Multiway Study).

- **55% Multiway Flop Rate in Tracked Live Cash (Hand2Note Live Database):** Across 239,000 postflop hands, only **45% (109k hands) are heads-up (2 players)**, whereas **24% (71k hands) are 3-way** and **31% (59k hands) involve 4 or more players**. This yields an empirical mean of **~2.95 to 3.25 players per flop** overall, rising to **3.2 to 4.2 players per flop** in soft $1/$2–$1/$3 games and zero-rake home games **[Judgement]**.
- **Flop Player Count by Preflop Action Line (PokerTube $1/$2 Benchmarks):** When the big blind checks an unraised limped pot, an average of **3 to 5 players** see the flop (pot = $6–$10 / 3–5 bb); when players over-limp behind in late position, **3 to 6 players** see the flop (pot = $8–$14 / 4–7 bb); when a player executes a 4x–5x + 1/limper isolation raise, the field narrows to an average of **2 to 3 players** (pot = $20–$35 / 10–17.5 bb).
- **Multiway 3-Bet Pots:** When a TAG player 3-bets in live cash, **all opponents fold preflop only 26% of the time** (meaning **74% of 3-bets see further action**), and **in 28% of cases a player behind responds with a cold-call or 4-bet** (Hand2Note Live Database).

1.4 Stack-Depth Distribution (Including 30–60 bb Buy-Ins) & Effective SPR Compression

Unlike online cash games where auto-top-up locks effective stacks at a uniform 100 bb, low-stakes live and home games operate across three distinct stack-depth tiers—and suffer from severe **effective SPR compression** caused by oversized preflop raises. Specifically, a 100 bb ($200–$300) live stack facing a 5x–6x bb open enters a 3-way flop with an **SPR of 4.8–5.2**, which is mathematically identical to the **35–50 bb stack depth** of Riverline's target home game facing a 2.5x–3.5x bb open (SplitSuit Live vs. Online, PokerCoaching 50 bb Review) **[Judgement]**:

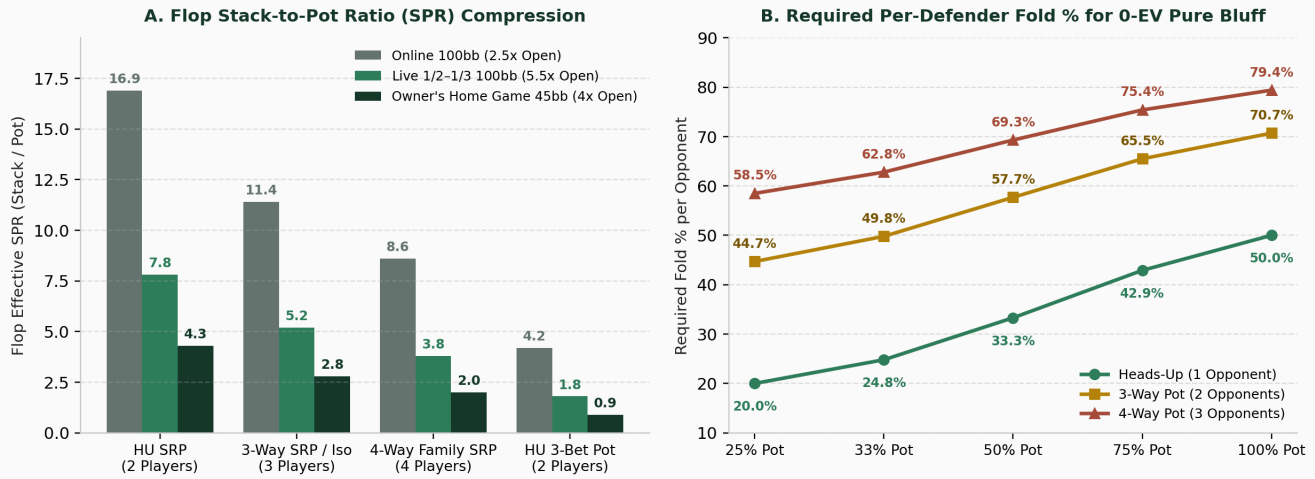


Figure 2: (A) Effective Flop Stack-to-Pot Ratio (SPR) comparison showing why a 100 bb live stack facing a 5.5x bb open plays at a compressed SPR (5.2 in 3-way SRP; 1.8 in HU 3-bet pot) close to Riverline's 45 bb home-game target (SPR 2.8 in 3-way SRP; 0.9 in 3-bet pot) **[Judgement]**. (B) Per-opponent break-even fold percentage required for a 0-EV pure bluff across 1, 2, and 3 opponents (GTO Gecko Multiway Study).

Stack-Depth Tier	Nominal Buy-In & BB Range	Published Venue & Home-Game Evidence	Flop SPR & Strategic Consequences
1. Short / Cap-Constrained Tier	**20–50 bb** ($40–$100 at $1/$2; $100 min at $1/$3 = **33.3 bb**)	• **US Casino Minimums:** $100 (**33.3 bb**) is the universal minimum buy-in for $1/$3 at Aria, Bellagio, Wynn, Caesars, Venetian, and Horseshoe (r/poker 1/3 Short-Stack Study). • **California Caps:** Cardrooms cap $1/$2 max buy-ins at **40–50 bb** ($40–$100 max at Players Casino Ventura, Commerce, and SoCal $1/$2 rooms).	With a $12–$15 open and 2 callers (pot = 15–20 bb), a 33–50 bb stack hits the flop with **SPR 1.0–2.5**. Set-mining (requiring 15:1–20:1 implied odds, Quora Stack Depth Analysis) is unprofitable; high-card top-pair equity and 3-bet jams dominate.
2. Home-Game & Recreational Mid Tier	**30–60 bb** (Owner's 6–8 handed no-rake home game; $100–$180 at $1/$3)	• **Owner's Primary Home Game (Brief §1):** 6–8 handed, **30–60 bb buy-ins**, zero rake. • **Coaching Benchmark:** Jonathan Little's Master the Fundamentals uses **50 bb** as the standard recreational cash effective stack, as casual players sit with 30–60 bb to keep 2–3 bullets per session **[Judgement]**.	At 45 bb with a 4x bb open: **HU SRP SPR = 4.3**; **3-Way SRP SPR = 2.8**; **HU 3-Bet (to 12 bb) SPR = 0.9–1.4**. Because HU 3-bet pots have SPR ≤ 1.5, b-inary/postflop-solver solves exact postflop leaf EVs in seconds (Brief §5) **[Judgement]**.

Stack-Depth Tier	Nominal Buy-In & BB Range	Published Venue & Home-Game Evidence	Flop SPR & Strategic Consequences
3. Full Buy-In & Deep / Straddled Tier	**80–167 bb** standard; **200–1,000+ bb** deep/straddled	• **Full Casino Caps:** $300 (100 bb) at Caesars, Venetian, Horseshoe $1/$3; $500 (167 bb) at Aria, Bellagio, Wynn $1/$3; $1,000 (333 bb) at MGM Grand $1/$3. • **Straddles:** Hand2Note's Live Database shows **85% of lower-stakes games include a 2x bb straddle**, halving effective stacks.	**Hidden SPR Compression:** A 100 bb ($300) live $1/$3 stack facing a $16.50 (5.5x bb) open has the exact same postflop SPR (**5.2 in a 3-way SRP**) as a **45 bb stack** facing a 2.5x bb open. Late home-game rebuys create asymmetric stacks (35 bb vs. 220 bb) **[Judgement]**.

1.5 Common Mistakes of Recreational Players in Live & Home Games

1. **Playing 2x–3x Too Many Hands Preflop (Especially Out of Position & Dominated Offsuit Broadways):** Recreational players enter 35%–50%+ of pots by limping and cold-calling hands like KJo, QTo, A7o, and weak suited gappers that suffer severe reverse implied odds multiway (SplitSuit Live Cash Truths, r/poker Live 1/2 Flatting Study). As SplitSuit notes, 95% of live players would immediately improve by folding half of their current preflop range.
2. **Inelastic Preflop & Flop Calling Regardless of Sizing or Effective Stack ("Set-Mining at 35 bb"):** Loose-passive players routinely limp-call 5x–7x bb isolation raises at 30–60 bb effective stacks with small pairs (22–55) or suited connectors, violating the 15:1–20:1 implied-odds threshold required to set-mine profitably (SplitSuit Live vs. Online, Quora Stack Depth Analysis, PokerCoaching 50 bb Review).
3. **Ignoring "Multiway Responsibility" & Overvaluing One-Pair / Non-Nut Draws:** In 3- to 5-way pots, probability that at least one opponent holds a pair or better on K♠ 7♥ 2♦ jumps from **35.8% (1 opponent) to 69.6% (2 opponents) and 82.0% (3 opponents)**, while opponent two-pair+ frequency quadruples from **2.4% to 11.1%** (GTO Gecko Multiway Study). Recreational players routinely call down or stack off 100 bb with weak top pairs, second pairs, or 8-high flush draws multiway (SplitSuit Live vs. Online, Crush Live Poker Glossary).
4. **Face-Up Bet-Sizing Tells & OOP "Blocker/Feeler" Donk Bets:** Unstudied players frequently lead out of position (donk-bet) for 15%–33% pot ("throwing out a $5 or $10 chip") with weak top pair, middle pair, or draws to "see where they stand" and try to set their own price, while reserving large 75%–100%+ pot bets and check-raises exclusively for two pair or better (SplitSuit Live vs. Online, TightPoker Range Analysis, r/poker OOP Donk Bet Thread).
5. **Extreme Turn/River Passivity & Massive Under-Bluffing in Raised Lines:** While recreational players over-call across all streets, their preflop 3-bet/4-bet ranges (often strictly QQ+/AK or KK+) and turn/river raise ranges are severely under-bluffed (SplitSuit Live Cash Truths, r/poker Low-Stakes Leaks). Conversely, when acting as preflop raiser, casual players either fire one "aimless c-bet" with Ace-high on the flop and give up, or check back strong one-pair hands on the river and miss thin value (Quora 1/2 Adjustment Guide, Crush Live Poker Glossary).
6. **Misapplying "Piecemeal Online GTO" in Multiway Live Games:** Semi-studied players lose heavily by copying 25%–33% heads-up range c-bets into 4-way live pots or bluff-catching rivers using Minimum Defense Frequency (MDF) against opponents who bluff 5% instead of the equilibrium 30% (TwoPlusTwo Live GTO Wizard Thread, Upswing Exploitative Deviations).

1.6 Master Comparison: Online 6-Max / 9-Max vs. Live $1/$2–$1/$3 vs. Owner's 30–60 bb Home Game

Dimension	Online 6-Max / 9-Max (100 bb Raked, Brief §1)	Live $1/$2–$1/$3 Casino (8–9 Handed)	Owner's 30–60 bb Home Game (6–8 Handed, No Rake, Brief §1)
Hands per Hour	**60–100/hr** per table; **200–250/hr** fast-fold (SplitSuit, Hand2Note).	**20–30 hands/hr** (~1 hand every 2.5–3 min) (SplitSuit, PrimeDope).	**15–25 hands/hr** (manual dealing, social pace, recurring friends) **[Judgement]**.
Recreational Density	**0–1 recreational player** per table (Hand2Note).	**90%** of tables have ≥2 fish; **57%** have **3–5+ fish** (Hand2Note).	**70%–100% recreational / semi-studied** with stable personal habits (SplitSuit) **[Judgement]**.
Open-Limp Frequency	**~0%** outside SB (PokerTube, Red Chip Poker).	**55%–75%** of unraised hands feature ≥1 open-limp (PokerTube) **[Judgement]**.	**60%–80%** of unraised hands feature open-limps / family pots **[Judgement]**.
RFI & Iso Sizing	**2.0x–2.5x bb** RFI; **3.5x + 1/limp** iso (Upswing, PokerTube).	**4x–7.5x bb** RFI ($10–$20); **4x–5x + 1/limp (6x–10x bb)** iso (SplitSuit, PokerTube).	**3x–5x bb** RFI; **4x–6x bb** iso; frequent non-standard 2x or 7x sizes **[Judgement]**.
Multiway Flop Share	**12%–20%** multiway; **80%–88%** heads-up (GTO Gecko).	**55% multiway** (**45%** HU, **24%** 3-way, **31%** 4+ way) (Hand2Note).	**65%–80% multiway** (zero rake removes preflop flatting penalty) **[Judgement]**.
3-Bet Response	**55%–65%** fold to 3-bet; ~0% cold-call 3-bet (Hand2Note).	Only **26%** all-fold to 3-bet; **28%** cold-call or 4-bet behind (Hand2Note).	Low fold to 3-bet; at 30–60 bb, 3-bets commit stacks (SPR 0.8–1.5) **[Judgement]**.
Rake & Drop Model	**4.5%–5% capped** (penalizes small flats) (SplitSuit).	**10% to $5–$6 + $1–$2 promo** (or $5–$8 flat flop drop in CA) (SplitSuit, r/poker).	**0% (No Rake)** in home game (or 0.1 bb/seat in ClubGG rules) (Brief §1).

2. Ranked List of the 20 Most Common Spot Types in Home & Low-Stakes Live Cash Games

Combining the Hand2Note Live Database flop distribution (45% HU, 24% 3-way, 31% 4+ way), PokerTube's $1/$2 preflop response metrics, and the owner's 6–8 handed 30–60 bb no-rake home-game structure (`RIVERLINE_BRIEF.md` §1), the table below ranks the **20 most frequent preflop and flop decision spots** by per-orbit incidence and maps each spot to its **Heuristic Baseline Difficulty** and honest **Riverline Provenance Tier** (Brief §2, Principle 1) **[Judgement]**:

Rank	Spot Type (Street · Action Configuration)	Est. Street Share [Judgement]	Flop SPR (45 bb Home vs. 100 bb Live)	Heuristic Difficulty [Judgement]	Riverline Provenance Tier (Brief §2.1 & §5) [Judgement]
1	**Preflop — Facing 1 Open-Limper (MP/CO/BTN/SB):** Isolation Raise (4x–5x+1) vs. Over-Limp vs. Fold	18%–22% of preflop	3.0–4.5 (45 bb) 7.0–9.0 (100 bb)	**Extreme**	**Estimated** now → **Modeled vs. Limp Profile** (M2, never fake "GTO")

Rank	Spot Type (Street · Action Configuration)	Est. Street Share [Judgement]	Flop SPR (45 bb Home vs. 100 bb Live)	Heuristic Difficulty [Judgement]	Riverline Provenance Tier (Brief §2.1 & §5) [Judgement]
2	**Preflop — Facing 2+ Limpers ("Limp-Train" in CO/BTN/SB):** Large Value Iso (6x–8x) vs. Over-Limp Speculative vs. Fold	14%–18% of preflop	1.5–2.8 (45 bb) 3.5–5.5 (100 bb)	**Extreme**	**Estimated** (Off-tree multi-limp node; Exact pot odds & call price)
3	**Preflop — Unopened Pot RFI (EP/MP/CO/BTN/SB):** Open-Raise Sizing & Range Width with Sticky Callers Behind	14%–16% of preflop	2.8–4.5 (45 bb) 5.5–8.5 (100 bb)	**Medium**	**Estimated** now → **Verified (Fixed-Tree Equilibrium)** in M2/M3
4	**Preflop — Facing a Large (4x–6x bb) Open Raise (MP/CO/BTN/SB):** Cold-Call vs. Linear 3-Bet vs. Fold	10%–13% of preflop	2.8–4.0 (45 bb) 6.5–8.5 (100 bb)	**Hard**	**Estimated** now → **Verified** (with SPR-adjusted implied-odds floor)
5	**Preflop — Facing an Open Raise + 1–2 Cold-Callers (CO/BTN/SB/BB):** Multiway Over-Call vs. Squeeze vs. Fold	9%–12% of preflop	1.8–3.0 (45 bb) 4.0–6.0 (100 bb)	**Extreme**	**Estimated** (Requires multiway EQR & squeeze fold-equity warning)
6	**Preflop — Big Blind Option vs. 1–4 Limpers:** Free Check to Multiway Flop vs. OOP Isolation Raise (5x+1/limper)	8%–10% of preflop	7.0–15.0 (check) 1.5–3.0 (iso 45 bb)	**Hard**	**Exact** free-check rule; **Estimated** OOP iso-raise threshold
7	**Preflop — Big Blind Defense vs. Single Open Raise (Closing Action Heads-Up vs. 3.5x–6x bb Open):** Call vs. 3-Bet vs. Fold	7%–9% of preflop	3.2–5.5 (45 bb) 7.5–11.5 (100 bb)	**Medium**	**Exact · pot odds** + **Verified** (via `b-inary/postflop-solver` leaf EVs)
8	**Flop — Multiway SRP / Iso-Pot (3–4 Players) as Preflop Aggressor IP:** Small/Medium C-Bet vs. Check Back	15%–18% of flop	1.5–3.5 (45 bb) 4.0–7.5 (100 bb)	**Extreme**	**Estimated** (Must apply multiway c-bet brake & compound fold math)
9	**Flop — Multiway SRP / Iso-Pot (3–4 Players) as Preflop Aggressor OOP:** Check to Field vs. Value C-Bet	12%–15% of flop	1.5–3.5 (45 bb) 4.0–7.5 (100 bb)	**Extreme**	**Estimated** (High check frequency required when sandwiched OOP)
10	**Flop — Multiway Unraised Limped Pot (3–5 Players, BB Checked):** Checked Around to Late Position Stab vs. Check	11%–14% of flop	7.0–15.0 (45 bb) 16.0–30.0 (100 bb)	**Hard**	**Estimated** (Capped multi-limp ranges; non-equilibrium tree)
11	**Flop — Defending as Caller (IP or Sandwiched OOP) vs. C-Bet in a 3–4 Way SRP:** Call vs. Raise vs. Fold	10%–12% of flop	1.5–3.5 (45 bb) 4.0–7.5 (100 bb)	**Extreme**	**Estimated** (Strict nut-draw & top-pair kicker filtering required)

Rank	Spot Type (Street · Action Configuration)	Est. Street Share [Judgement]	Flop SPR (45 bb Home vs. 100 bb Live)	Heuristic Difficulty [Judgement]	Riverline Provenance Tier (Brief §2.1 & §5) [Judgement]
12	**Flop — Heads-Up SRP / Iso-Pot as IP Preflop Aggressor vs. BB / Limp-Caller Check:** C-Bet vs. Check Back	9%–11% of flop	3.0–5.5 (45 bb) 7.5–11.5 (100 bb)	**Easy–Med**	**Verified** in M2/M3 (Direct 2-player `binary/postflop-solver` spot)
13	**Flop — Heads-Up SRP / Iso-Pot as OOP Caller (BB or Limp-Caller) vs. IP C-Bet:** Check-Call vs. Check-Raise vs. Fold	8%–10% of flop	3.0–5.5 (45 bb) 7.5–11.5 (100 bb)	**Medium**	**Verified** in M2/M3 (Direct 2-player `binary/postflop-solver` spot)
14	**Preflop — Limper Facing an Isolation Raise Behind (EP/MP):** Limp-Call vs. Limp-Fold vs. Limp-Reraise/Jam	5%–7% of preflop	1.8–4.0 (45 bb) 4.5–8.0 (100 bb)	**Hard**	**Exact · incremental call price**; **Estimated** range continuation
15	**Flop — Facing an OOP Recreational Donk Bet (Small 20%–33% "Feeler" vs. Large 75%–100% Pot Lead):** Raise vs. Call vs. Fold	6%–8% of flop	2.0–4.5 (45 bb) 5.0–9.0 (100 bb)	**Extreme**	**Estimated** (Sizing-conditioned opponent range split required)
16	**Preflop — Opener Facing a 3-Bet at 30–60 bb vs. 100 bb+:** 4-Bet Shove vs. Call vs. Fold	4%–6% of preflop	0.6–1.5 (45 bb) 2.2–3.8 (100 bb)	**Hard**	**Verified** at 30–60 bb (Low-SPR `binary/postflop-solver` + jam oracle)
17	**Flop — Heads-Up SRP as OOP Preflop Aggressor (EP/MP) vs. IP Cold-Caller (CO/BTN):** C-Bet vs. Check	5%–7% of flop	3.0–5.0 (45 bb) 7.5–10.5 (100 bb)	**Medium**	**Verified** in M2/M3 (Direct 2-player `binary/postflop-solver` spot)
18	**Preflop — Small Blind vs. Big Blind (Folded to SB):** Limp-Complete vs. Raise (3x–4x bb) vs. Fold	3%–5% of preflop	5.5–18.0 (45 bb) 12.0–45.0 (100 bb)	**Easy–Med**	**Verified** in M2/M3 (2-player BvB subgame equilibrium)
19	**Flop — 3-Bet Pot (Heads-Up or 3-Way Cold-Called) at Low SPR (SPR 0.6–2.5):** Commitment C-Bet / Jam vs. Check	4%–6% of flop	0.5–1.5 (45 bb) 2.0–3.5 (100 bb)	**Medium**	**Verified** for HU (SPR ≤ 2 solved exactly in `binary/postflop-solver`)
20	**Flop — Facing a Flop Raise or Check-Raise in a Single-Raised or Multiway Pot:** Continue vs. Fold Overpair/TPTK	3%–5% of flop	1.5–4.0 (45 bb) 4.0–8.0 (100 bb)	**Extreme**	**Estimated** (Must warn on heavily nutted recreational raise ranges)

3. Diagnostic Analysis: Which Spots Are Hardest for Riverline's Heuristic Baseline (and Why)

Riverline currently relies on a deterministic heuristic baseline (marked `Estimated` per `RIVERLINE_BRIEF.md` §5) that combines canonical hand evaluation, pot-odds/call-price thresholds, and conditional Monte Carlo showdown sampling against assumed opponent ranges. While this baseline performs reliably in **heads-up single-raised and 3-bet pots** (Ranks **#3, #7, #12, #13, #17, #18, #19**), it faces severe mathematical distortion in **four specific home/live spot clusters [Judgement]**:

3.1 Failure Mode 1: Multiway Flop C-Betting & Defense (Ranks #8, #9, #11) — Non-Linear Equity Realization & Compound Fold-Equity Collapse

- **Empirical Solver Proof:** In GTO Wizard's 3-way postflop benchmark (LJ opener vs. SB and BB callers), adding a single second caller increases the opener's flop checking frequency by **+11%** and collapses large (100% pot) c-bet usage from **18% (heads-up) down to 1.3% (3-way)**. Furthermore, when the preflop raiser is sandwiched between a caller in position and the big blind (e.g., CO opens, BTN calls, BB calls — GTO Wizard 3-Way Probing Study), the solver checks CO's *entire range (~100%)* on almost all flop textures because BTN acts behind.
- **Compound Bluff Math:** Under an independent opponent response model (GTO Gecko Multiway Solver Study), a 50%-pot pure bluff needs a single heads-up opponent to fold **33.3%** of the time, but requires each of two opponents in a 3-way pot to fold **57.7%** ($\sqrt{(1/3)}$) and each of three opponents in a 4-way pot to fold **69.3%** ($(1/3)^{1/3}$).
- **Why Riverline's Heuristic Baseline Struggles:** Monte Carlo all-in equity sampling on the flop measures raw showdown share (e.g., AQs has 31.6% 3-way equity on K♠ 7♥ 2♦ vs. 33.3% fair share, GTO Gecko), which tricks a naive heuristic into treating AQs or 99 (45.2% 3-way equity) as profitable flop c-bets or calls. In reality, because two opponents hold a pair or better **69.6%** of the time (and **82.0%** 4-way), non-nutted one-pair and ace-high hands suffer catastrophic reverse implied odds when betting or calling multiway **[Judgement]**.

3.2 Failure Mode 2: Preflop Iso-Raising vs. Over-Limping (Ranks #1, #2, #6) — Off-Tree Non-Equilibrium Nodes & Check-Down Bias

- **Why Nash Equilibrium Cannot Exist Unconditioned Here:** In any multi-player game with 3+ active entrants, a Nash equilibrium is neither unique nor guaranteed to maximize EV against human policies (GTO Gecko / CMU Pluribus Note). Moreover, because equilibrium open-limping in UTG–CO is 0% (Red Chip Poker), any hand with 1–4 limpers is an **off-tree node** that requires an explicit behavioral model of the limpers **[Judgement]**.
- **Why Riverline's Heuristic Baseline Struggles:** A heuristic that ranks hands by raw preflop equity over-isolates offsuit broadways (`KJo`, `ATo`, `QJo`) into 3–4 sticky limp-callers (where they get dominated by `AK`/`AQ`/`KQs` or outdrawn by two-pair/flushes) while under-overlimping suited wheel aces (`A2s–A5s`), suited kings (`K8s–KTs`), and small pairs (22–66) that generate massive nut-flush and set equity in cheap 4- to 5-way family pots (r/poker Sticky Limpers Study, r/poker Live 1/2 Flatting Study) **[Judgement]**.

3.3 Failure Mode 3: Facing Flop Check-Raises & OOP Donk Leads (Ranks #15, #20) — Path-Unconditioned Opponent Sampling

- **Empirical Population Asymmetry:** In low-stakes live and home games, a flop raise or check-raise (especially multiway) is almost never a balanced semi-bluff; it is overwhelmingly weighted toward **two pair, sets, or combo draws** (r/poker Casino Guide, r/poker Low-Stakes Leaks). Conversely, a small (20%–33% pot) OOP donk lead into the preflop raiser is typically a capped, vulnerable one-pair hand or weak draw trying to buy a cheap turn card (SplitSuit Live vs. Online).
- **Why Riverline's Heuristic Baseline Struggles:** Because Riverline's postflop heuristic samples opponent hands from a generic static range rather than a *path- and sizing-conditioned action model*, it gives a flop check-raiser far too many random one-pair/air combos (causing Hero to call off with doomed overpairs/TPTK) while respecting small OOP donk bets too much instead of raising them for value and protection **[Judgement]**.

3.4 Failure Mode 4: Preflop Cold-Calling & Set-Mining Against 4x–6x bb Opens at 30–60 bb (Ranks #4, #5, #14, #16) — The Implied-Odds Cliff

- **Mathematical Threshold:** Calling preflop strictly to hit a set (11.8% flop probability, ~7.5:1 against) requires **15:1 to 20:1 effective stack-to-call ratio** to account for boards where the set is not paid off or is over-setted (Quora Stack Depth Analysis).
- **Why Riverline's Heuristic Baseline Struggles:** In the owner's **30–60 bb home game** (or a $100 min buy-in at $1/$3 facing a $15 open), calling a 4x–5x bb raise costs **8% to 17% of the entire effective stack** (only 6:1 to 12:1 implied odds). If the heuristic evaluates preflop calls using 100 bb playability weights rather than computing `impliedOddsRatio = effectiveStackBb / callAmountBb`, it will recommend calling small pairs (22–55) and low suited connectors (65s, 54s) in spots where they are mathematically unprofitable (PokerCoaching 50 bb Review) **[Judgement]**.

3.5 Summary Matrix: Where the Heuristic Baseline Succeeds vs. Breaks Down

Across the top 20 home/live spots, Riverline's baseline reliability partitions cleanly into three operational zones **[Judgement]**:

- **High Baseline Fidelity (Ranks #3, #7, #12, #13, #17, #18, #19):** Heads-up single-raised pots, blind-vs-blind confrontations, and low-SPR 3-bet commitment spots (where raw equity, blocker counts, and pot-odds thresholds closely track 2-player b-inary/postflop-solver equilibria).
- **Moderate Baseline Fidelity Requiring SPR/Playability Gates (Ranks #4, #6, #10, #14, #16):** Single-raiser defense against 4x–6x bb live sizing and unraised checked-around limped flops, which require an explicit `effectiveStackBb / callAmountBb ≥ 15` implied-odds gate for small pairs and low suited connectors.
- **Low Baseline Fidelity Requiring Multiway & Path-Conditioned Rules (Ranks #1, #2, #5, #8, #9, #11, #15, #20):** Multi-limp isolation/over-limp decisions, 3- to 5-way flop c-betting/defense, and facing flop check-raises or OOP donk bets.

4. Architectural Extension: Translating Field Research into `RIVERLINE_BRIEF.md`

To serve the owner's **6–8 handed, 30–60 bb no-rake home game** and **9-max 100 bb+ online game** while honoring every non-negotiable principle in `RIVERLINE_BRIEF.md` (2026-10-21), the table below

specifies the concrete mathematical and product upgrades across Milestones **M1, M2, and M3 [Judgement]**:

Riverline Principle / Milestone (`RIVERLINE_BRIEF.md`)	**Architectural Requirement Grounded in Live / Home-Game Data [Judgement]**	**Cost, Licence & Risk Assessment (Brief §7)**
Principle 1: Honesty is the Brand (4 Tiers: `Exact`, `Verified`, `Estimated`, `Unavailable`; **no numeric confidence %**)	• Display **Exact · call price, pot odds & SPR** and **Exact · card blockers** (computed in `shared/poker-domain/`) on every spot. • Label 2-player HU SRP, BvB, and 30–60 bb 3-bet pots backed by offline b-inary/postflop-solver tables as **Verified · 2-player equilibrium**. • Label multiway (3+ player) and multi-limp spots as **Estimated · multiway heuristic** (or **Verified · vs. Loose-Passive Limp Model** when solved against an explicit RNR limper profile). Never display numeric confidence percentages (e.g., remove any legacy "83% confidence" copy).	**Cost:** $0 (in-house copy & provenance enum enforcement). **Effort:** 2–3 days in M1. **Risk:** Zero legal/IP risk; eliminates "fake GTO" complaints documented on TwoPlusTwo.
Principle 2 & M2: Unknown is Never Fold, Zero, or Excluded (Hand Reconstruction & Scenario Mode)	• In M2 Hand Reconstruction, allow a home-game player to enter a 4-way flop (e.g., *"2 early limpers, CO raised to 5 bb, BTN and limpers called, pot = 21.5 bb, effective stack = 45 bb"*) without forcing exact identities or hole cards for every folded seat. • Keep unanswered cells in Personal Strategy matrices in a first-class Unknown state rather than coercing them to `0% / Fold`.	**Cost:** $0. **Effort:** Built on existing `RANGE-CORE-001` / `SavedStudyObject` contracts. **Risk:** Prevents silent corruption of partial live hand memories.
Principle 3 & 5: Heuristic Disagreement is Never a "Mistake" (Personal Strategy vs. Baseline)	• Replace judgmental error flags with descriptive comparative badges: `Inside baseline mix`, `Close to baseline boundary`, `Outside baseline · matches your Home Game profile`. • Support named environment profiles (e.g., *"6–8p Home Game · 30–60 bb No Rake"* vs. *"9-Max Online · 100 bb"*) with 3 user-named modes so exploitative iso-raising and over-limping are stored cleanly as intentional strategy.	**Cost:** $0. **Effort:** Core M2 Personal Strategy workflow. **Risk:** Zero; directly solves the #1 live-player frustration with rigid online trainers.
M1 & M2 Heuristic Patch: Fixing the Top 4 Home-Game Failure Modes (In `shared/poker-domain/`)	1. **Implied-Odds / SPR Gate (Ranks #4, #5, #14):** Penalize set-mining (22–66) and low suited connectors when `effectiveStackBb / callAmountBb < 15` (critical at 30–60 bb facing 4x–6x opens). 2. **Multiway Flop C-Bet Brake (Ranks #8, #9, #11):** Scale flop c-bet frequency by $1 / k^{1.4}$ for k opponents, suppress 100%-pot c-bets multiway (GTO Wizard 3-Way Data), and enforce near-100% checking when sandwiched OOP. 3. **Action-Conditioned Postflop Sampling (Ranks #15, #20):** Weight opponent flop check-raise/big-lead samples heavily toward two-pair+ while weighting small (20%–33% pot) OOP donk leads toward weak pairs/draws.	**Cost:** $0 (pure math upgrades in canonical domain engine). **Effort:** 1–2 weeks. **Risk:** Must pass regression gates (~2,750 tests) and preserve monotonicity invariants.

Riverline Principle / Milestone (`RIVERLINE_BRIEF.md`)	Architectural Requirement Grounded in Live / Home-Game Data [Judgement]	Cost, Licence & Risk Assessment (Brief §7)
M3 Solver Expansion: Closing the Preflop & Multiway Gap Legally (Brief §5 & Principle 7)	• **Heads-Up & Low-SPR Bridge:** Use offline b-inary/postflop-solver (AGPL-3.0, shipping pure numerical output tables only) across a canonical 47-flop subset to compute exact postflop leaf EVs for 30–60 bb no-rake SRPs (SPR 2.8–4.5) and 3-bet pots (SPR 0.9–1.5). • **Multiway & Limp Blueprints:** Pair with MIT-licensed Rust MCCFR (conorarmstrong/pluribus / exinori/DCFR-SOLVER) extended to 6–9 seats with Restricted Nash Response (`--rnr-model`) locked to 2–3 transparent Recreational Limper archetypes.	**Cost:** $0 software license (MIT + offline AGPL numerical output tables; verify AGPL isolation before public release). **Effort:** 3–6 weeks in M2/M3. **Risk:** High multiway MCCFR variance — mitigate by presenting coarse action guidance rather than 1% mixed precision.

5. Risks, Limitations & Open Questions for the Owner

- **No Unconditioned "GTO" Exists for Multiway Limped Family Pots [Judgement]:** Because open-limping from early/middle position is 0% in a converged no-rake equilibrium, Riverline cannot honestly label any response to 2–3 home-game limpers as "Verified Equilibrium." Instead, Riverline should offer 2–3 transparent opponent assumptions (e.g., *"Capped Loose-Passive Limper (VPIP 40% / PFR 8%)"* vs. *"Trappy Limp-Reraiser (includes AA/KK)"*) and scope the badge as `Estimated · vs. Capped Limper` or `Verified · vs. Loose-Passive Model`.
- **Empirical Gap on Exact 30–60 bb No-Rake Home-Game Hand Databases (UNVERIFIED):** While Hand2Note's 500k-hand Live Database, GTO Gecko's multiway simulations, and PokerAtlas casino rules provide rigorous numbers for live casino games, no public million-hand optical/RFID database exists specifically for private 30–60 bb zero-rake kitchen-table games (**UNVERIFIED** in public literature). Riverline's own local-first Hand Reconstruction and Game Night logs (M2) will become a proprietary source of truth for the owner's exact group **[Judgement]**.
- **Three Decisions for the Owner:**
  1. *Default Home-Game Preset Parameters:* Should the built-in **Home Game** preset default to **7-handed, 45 bb effective stack, 0% rake, 4x bb standard open**, or do you prefer 6-handed / 8-handed and a specific starting buy-in (e.g., 40 bb or 50 bb)?
  2. *Limper Archetypes in Scenario/Playbook Mode:* When a user clicks "1 Limper" or "2+ Limpers" in Playbook/Analyze, should the baseline assume a **Capped Recreational Limper** by default, with a one-click toggle for **Trappy Limper (can hold AA/KK)**?
  3. *Multiway Postflop Granularity:* In 3- to 5-way flops (Ranks #8–#11), should Riverline simplify action outputs to 3 clear categories (`Check / Fold`, `Small Bet 33%`, `Value Bet 66%+`) to avoid false precision?

6. Complete Verified Source Registry

- **Hand2Note Live Poker Database (Andrii Plakhotnyi, July 2025):** Online vs. Live Poker: Insights from the Live Poker Database — 500,000+ live hands; flop player distribution (45% HU, 24% 3-way, 31% 4+ way across 239k flops), recreational table density (90% ≥2 fish, 57% 3–5 fish), 3-bet fold rate (26% all-fold, 28% cold-call/4-bet), straddle incidence (85% at lower stakes), TAG VPIP/PFR (31/20, WWSF 40%, Aggression 1.6).

- **GTO Gecko Multiway Solver Study (Aug–Sept 2026):** Multiway Pots in Poker: 5 Rules Backed by Solver Data — 400,000 runouts/cell on K♠ 7♥ 2♦ showing HU vs. 3-way vs. 4-way showdown equity compression (77, AK, KQ, 99, AQs), opponent pair+ probabilities (35.8% → 69.6% → 82.0%), and compound break-even bluff fold rates.
- **GTO Wizard 3-Way Postflop Solver Benchmarks (March 2026):** Playing In Position Against Two Callers (+11% check frequency, 100%-pot c-bet drop from 18% to 1.3%, 9.8% SB donk lead on low boards) and Probing in 3-Way Pots (near-100% check frequency by sandwiched opener).
- **PokerTube Preflop Limping & $1/$2 Pot Benchmarks (Jan 2026):** What Is Limping in Poker? Why It Shapes Pots Preflop — $1/$2 iso-raise formula (4x bb IP / 5x bb OOP + 1 bb/limper), average players to flop (3–5 on BB check, 3–6 on over-limp, 2–3 on iso-raise), and preflop pot sizes ($6–$35).
- **SplitSuit / Red Chip Poker Live Cash Benchmarks (2025–2026):** Live Poker Cash Game Truths In 2026, Poker Live Versus Online: 6 Differences, and Red Chip Poker Preflop Charts — 20–30 hands/hr live vs. 60–100 online, 5x–7x bb live open sizes, 10% rake ($5+$1/$6+$2 cap), and recreational leak profiles.
- **PokerAtlas & California / Las Vegas Buy-In Directories (2026):** PokerAtlas Las Vegas Cash Game Directory ($100 min = 33.3 bb at $1/$3 across Aria, Bellagio, Wynn, Caesars, Venetian, Horseshoe), Players Casino Ventura 50 bb Cap Review, and PokerCoaching 50 bb Cash Course Review.
- **Crush Live Poker, Upswing Poker & Community Field Reports:** Crush Live Poker Glossary (Multiway Responsibility, Attacking Limped Pots), Upswing 6-Max Guide, Upswing Exploitative Deviations, PokerNews vs. Limpers Solver Study, TwoPlusTwo Live Cash GTO Thread, and r/poker Sticky Limpers Study.